

Sungmin Yun

Computer Architectures and Systems for AI

Email: sungmin.yun11@gmail.com

GitHub: github.com/sungmin-yun

Web: sungminyun.me

Phone: +82-10-2745-7538

Education

Seoul National University, Postdoctoral Researcher

Sep 2025 – Present

Seoul National University, Ph.D. in Artificial Intelligence

GPA: 4.1/4.3 Aug 2020 – Aug 2025

Yonsei University, B.S. in Integrated Information Technology

GPA: 3.6/4.3 Mar 2017 – Feb 2020

Summary

My current work focuses on **designing efficient computer architectures and serving systems for large-scale AI models** through algorithm–hardware co-design and system-level optimization. To this end, I continuously track the latest trends in both AI models and hardware developments.

My expertise lies in building computer architectures that efficiently execute diverse AI models. I have developed specialized accelerators for Large Language Models, Graph Neural Networks, and Recommender Systems, which has provided me with deep insight into the computational behavior of modern AI workloads. In particular, I have investigated emerging AI models such as Mixture-of-Experts and Multi-Head Latent Attention (MLA), analyzing their distinct compute and memory characteristics to guide hardware–software co-design.

Publications

[arXiv 2025] The New LLM Bottleneck: A Systems Perspective on Latent Attention and Mixture-of-Experts, **S. Yun**, S. Park, H. Nam, Y. Lee, G. Lee, K. Kyung, S. Kim, N. S. Kim, J. Kim, H. Kim, J. Cho, S. Baek, J. H. Ahn

[IEEE CAL 2025] SSD Offloading for LLM Mixture-of-Experts Weights Considered Harmful in Energy Efficiency, K. Kyung, **S. Yun**, J. H. Ahn

[IEEE CAL 2025] COSMOS: A CXL-Based Full In-Memory System for Approximate Nearest Neighbor Search, S. Ko, H. Shim, W. Doh, **S. Yun**, J. So, Y. Kwon, S. Park, S. Roh, M. Yoon, T. Song, J. H. Ahn

[HPCA 2025] Anaheim: Architecture and Algorithms for Processing Fully Homomorphic Encryption in Memory, J. Kim, **S. Yun**, H. Ji, W. Choi, S. Kim, J. H. Ahn

[MICRO 2024] Duplex: A Device for Large Language Models with Mixture of Experts, Grouped Query Attention, and Continuous Batching, **S. Yun**, K. Kyung, J. Cho, J. Choi, J. Kim, B. Kim, S. Lee, K. Sohn, J. H. Ahn

[ICS 2024] CLAY: CXL-based Scalable NDP Architecture Accelerating Embedding Layers, **S. Yun**, H. Nam, K. Kyung, J. Park, B. Kim, Y. Kwon, E. Lee, J. H. Ahn

[IEEE TC 2023] GraNDe: Efficient Near-Data Processing Architecture for Graph Neural Networks, **S. Yun**, H. Nam, J. Park, B. Kim, J. H. Ahn, E. Lee

[IEEE CAL 2022] GraNDe: Near-Data Processing Architecture With Adaptive Matrix Mapping for Graph Convolutional Networks, **S. Yun**, B. Kim, J. Park, H. Nam, J. H. Ahn, E. Lee

[MICRO 2021] TRiM: Enhancing Processor-Memory Interfaces with Scalable Tensor Reduction in Memory, B. Kim, J. Park, **S. Yun**, E. Lee, M. Rhu, J. H. Ahn

Experience

Postdoctoral Researcher , Seoul National University	Sep 2025 – Present
Samsung Electronics Internship	Mar 2023 – Aug 2023
Mandatory Military Service as Professional Research Personnel	Sep 2023 – Present (expected Aug 2026)

Awards

Best Ph.D. Dissertation Award Seoul National University	Aug 2025
Best of CAL Award 29th IEEE International Symposium on High-Performance Computer Architecture	Feb 2023

Invited Presentations

Poster presentation at Samsung AI Forum	Nov 2023
Oral presentation on Best of CAL session at IEEE International Symposium on High-Performance Computer Architecture	Feb 2023

Paper Review

IEEE Transactions on Computers
IEEE Transactions on Circuits and Systems for Artificial Intelligence

Skills

C/C++, Python, PyTorch, SGLang, vLLM, CUDA, Verilog